Predicate Logic

Equivalences without tables, and what happens when a statement has a variable in it.

Greg Ryslik · The Ohio State University

$\forall x\ P(x)$

Today

AGENDA

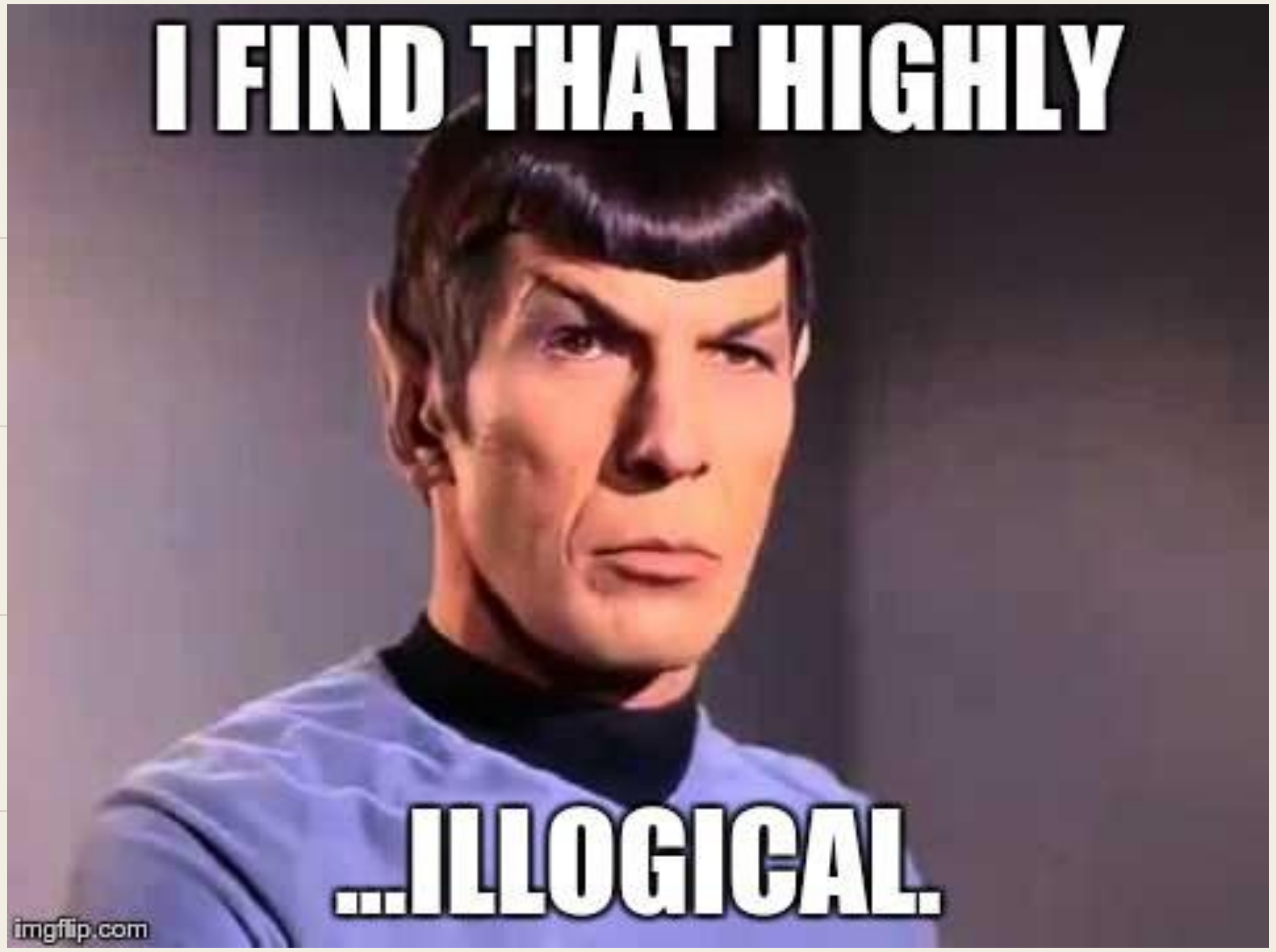
01 Tautologies and contradictions

02 Proving an equivalence without drawing a table

03 Predicates: propositions with a variable

04 The quantifiers $\forall$, $\exists$, $\exists!$

05 Negating a quantified statement



Tautology and Contradiction

DEFINITION

A compound proposition that is always true, no matter what the truth values of its variables, is a **tautology** . One that is always false is a **contradiction**.

TAUTOLOGY

$p \lor \neg p$

CONTRADICTION

$p \land \neg p$

NEITHER

$p \lor q$

Example: A Tautology by Table

Show that

$$(p \land q) \to (p \lor q)$$

is a tautology.

Four rows, every one true. Done — but notice you had to fill in six cells to justify four.

p	q	$p \land q$	$p \lor q$	$(p \land q) \to (p \lor q)$
T	T	T	T	T
T	F	F	T	T
F	T	F	T	T
F	F	F	F	T

Logical Equivalence

DEFINITION

Compound propositions p and q are **logically equivalent** if $p \leftrightarrow q$ is a tautology. We write $p \equiv q$.

$\equiv$ is not an operator. You cannot write it inside a formula.

It is a statement *about* two formulas: they agree in every row.

Example: A Tautology Without a Table

$[p \land (p \to q)] \to q$	show this is a tautology
$\equiv [p \land (\neg p \lor q)] \to q$	Implication law
$\equiv [(p \land \neg p) \lor (p \land q)] \to q$	Distributive law
$\equiv [F \lor (p \land q)] \to q$	Negation law
$\equiv (p \land q) \to q$	Identity law
$\equiv \neg(p \land q) \lor q$	Implication law
$\equiv (\neg p \lor \neg q) \lor q$	De Morgan's law
$\equiv \neg p \lor (\neg q \lor q)$	Associative law
$\equiv \neg p \lor T$	Negation law
$\equiv \mathbf{T}$	Domination law

This one has a name: **modus ponens**. If p holds and $p \to q$ holds, then q holds.

Why Tables Don't Scale

Variables	Rows
2	4
5	32
10	1,024
20	1,048,576
64	more than atoms you can name

2^n

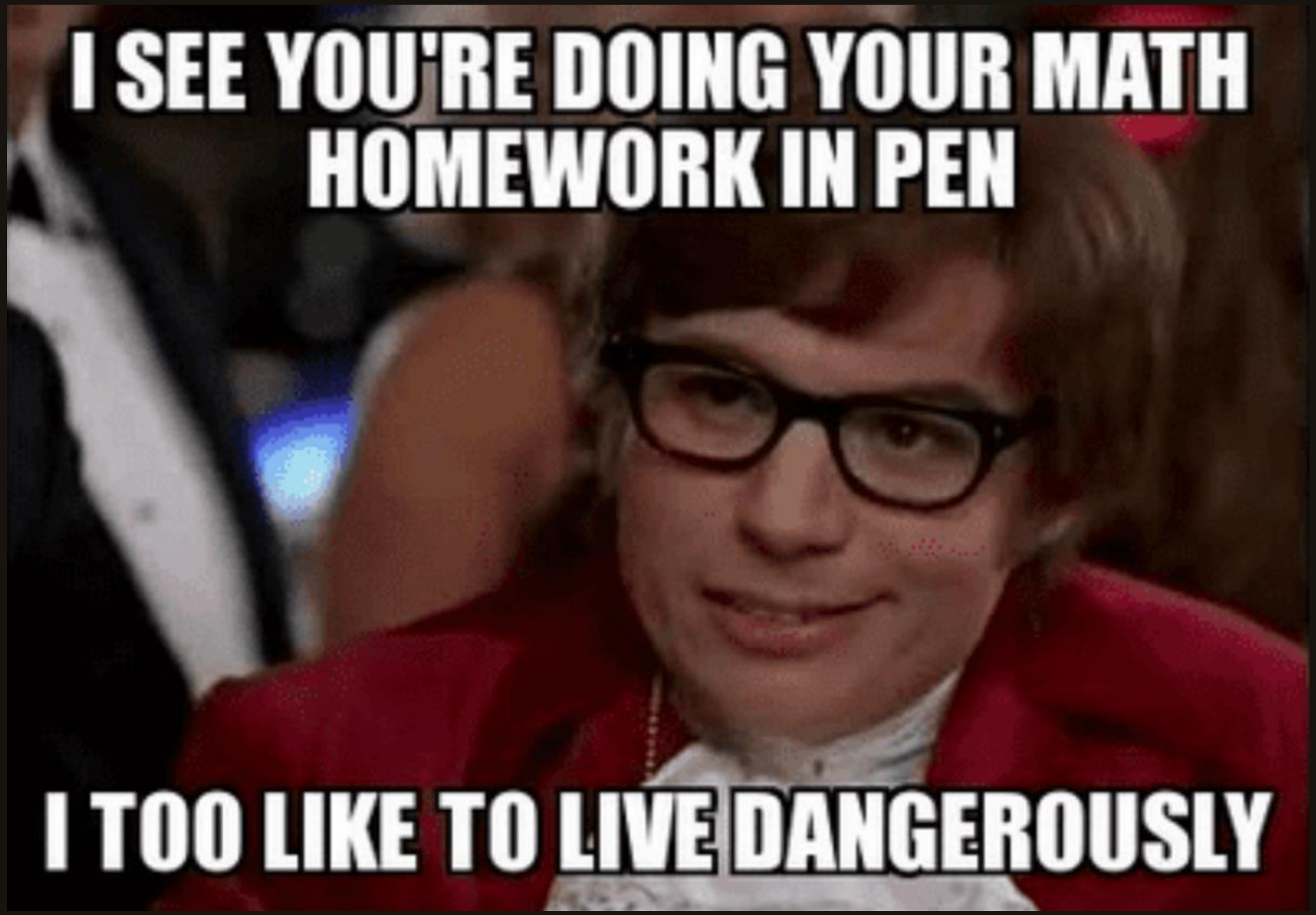
Every variable you add doubles the table. The laws do not care how many variables there are.

Deciding whether a formula can ever be true is the SAT problem — the first problem ever shown to be NP-complete.

HOMEWORK 1, HOUR 3

“I’ll just draw one more table.”

Row 512 of 1,024. There is a faster way, and it fits on one line.



Predicates

Are these true or false?

01	x is greater than 3.
02	Computer x is functioning properly.

Depends on x.

Neither is a proposition. Both become one the moment you fix `x`.

The Two Parts of a Predicate

`P(x) : x is greater than 3`

SUBJECT

The variable `x` — what the sentence is about.

PREDICATE

“is greater than 3” — a property the subject may or may not have.

VALUE AT 4

`P(4)` is a proposition, and it is true.

`P` is a **propositional function**: give it a value, get a proposition back.

Quantification

Assign a value to the variable and the predicate becomes a proposition. That is one way.

The other way is **quantification** : state how far across a domain the predicate holds.

English already does this, with words like *all*, *some*, *many*, *none* , and *few*. Mathematics keeps two of them.

$\forall$ holds for **every** element

$\exists$ holds for **at least one** element

The Universal Quantifier

DEFINITION

The **universal quantification** of P(x) is the statement "P(x) for all values of x in the domain," written $\forall x; P(x)$.

Read: *for all x, P(x)* — or *for every x*.

An element where P(x) is false is a **counterexample**.

One counterexample is enough. You never need a second.

The Existential Quantifier

DEFINITION

The **existential quantification** of P(x) is the statement "there exists an element x in the domain such that P(x)," written ∃x; P(x).

Read: *for some x*, *for at least one x*, or *there is an x* .

A single element that works is a **witness**.

Producing the witness is usually the whole proof.

The Uniqueness Quantifier

$$\exists!x;\ P(x)$$

“There exists a unique x such that $P(x)$ is true.”

OBLIGATION 1

Show at least one element works.

OBLIGATION 2

Show no second element does.

Half the proof is not the proof.

Quantifiers as Loops

Logic	Python
`∀x ∈ D; P(x)`	`all(P(x) for x in D)`
`∃x ∈ D; P(x)`	`any(P(x) for x in D)`
`¬(∀x; P(x))`	`not all(...) == any(not ...)`

Over a finite domain a quantifier is a loop with an early exit.

`all` breaks on the first false. `any` breaks on the first true. Counterexample and witness, in code.

Over an infinite domain the loop never returns. That is why we need the laws.

The Domain Changes the Answer

$$\forall x;\ (x^2 \geq x)$$

DOMAIN $\mathbb{Z}$

True

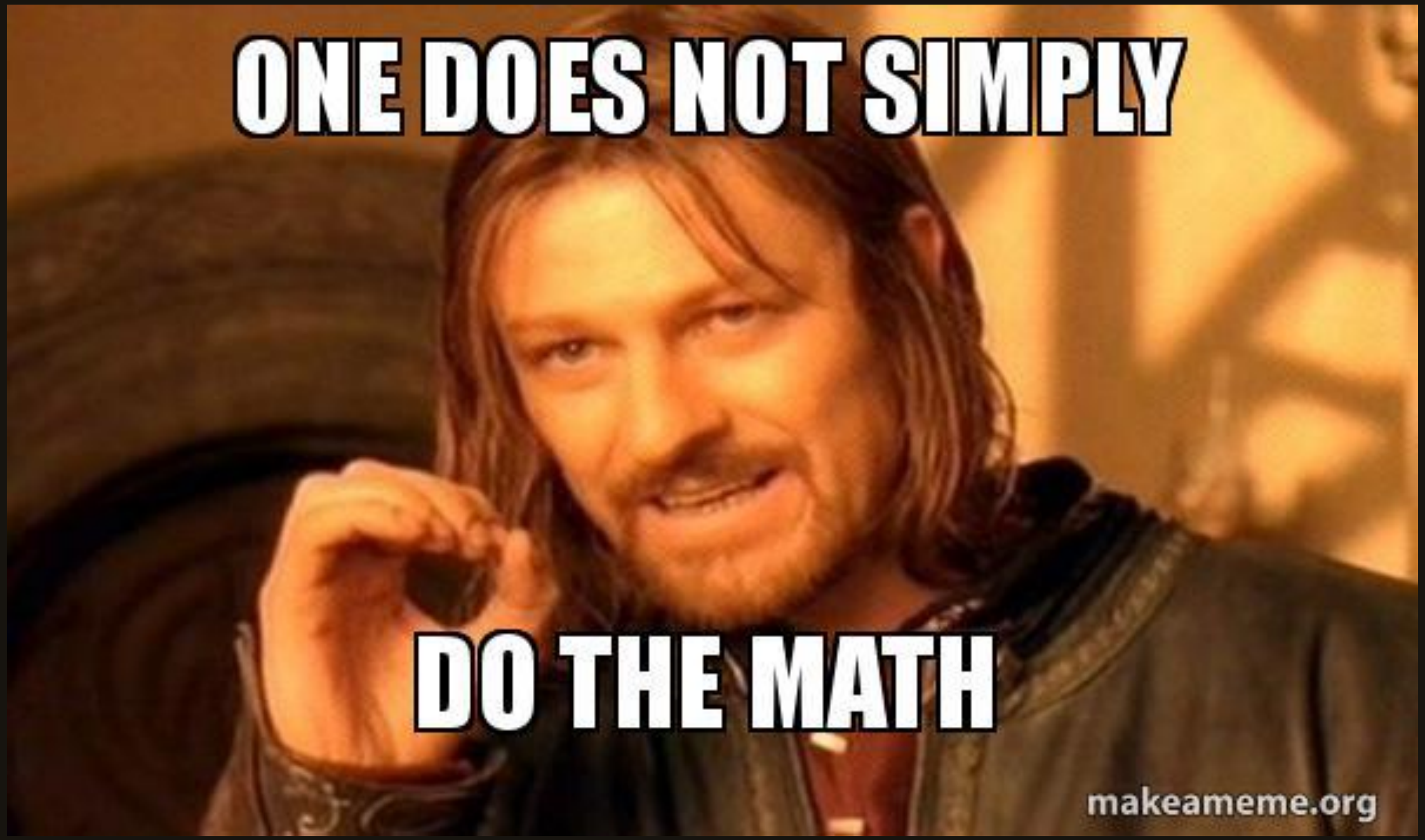
No integer sits strictly between 0 and 1, so squaring never shrinks.

DOMAIN $\mathbb{R}$

False

Take $x = 0.5$. Then $x^2 = 0.25 < 0.5$. One counterexample, done.

A quantified statement without a stated domain is not a statement.



∀ VS ∃

One does not simply check every real number.

To break a ∀ you need one example. To prove one you need an argument.

Example: Moving a Quantifier Past A

$$\exists x (A \to P(x)) \equiv A \to \exists x\ P(x)$$

where x does not occur free in A, and the domain is nonempty.

CASE 1 — A IS FALSE

Then $A \to P(x)$ is true for every x, vacuously. Since the domain has at least one element, the left side is true.

The right side is $F \to (\text{anything})$, also true. Both sides true.

CASE 2 — A IS TRUE

$$A \to P(x) \equiv \neg A \lor P(x) \equiv F \lor P(x) \equiv P(x)$$

So the left side is $\exists x\ P(x)$. The same rewrite on the right gives $F \lor \exists x\ P(x) \equiv \exists x\ P(x)$. Equal again.

Drop the nonempty assumption and case 1 breaks: over an empty domain the left side is false and the right side is true.

Quantifiers Over a Finite Domain

Let $D = \{x_1, x_2, \ldots, x_n\}$.

$$\forall x \in D;\ P(x) \equiv P(x_1) \wedge P(x_2) \wedge \cdots \wedge P(x_n)$$

$$\exists x \in D;\ P(x) \equiv P(x_1) \vee P(x_2) \vee \cdots \vee P(x_n)$$

$\forall$ is a long conjunction. $\exists$ is a long disjunction. Now negate both and see what De Morgan gives you.

Negating Quantifiers

$$\neg(\forall x;\ P(x)) \equiv \exists x;\ \neg P(x)$$

"Not everything works" means "something fails."

$$\neg(\exists x;\ P(x)) \equiv \forall x;\ \neg P(x)$$

"Nothing works" means "everything fails."

Push the negation inward, flip the quantifier. Same move as De Morgan — because it *is* De Morgan.

Example: Negating Statements About $\mathbb{R}$

NEGATE THESE

$\forall x \in \mathbb{R}; \ (x^2 > x)$

$\exists x \in \mathbb{R}; \ (x^2 = 2)$

$\neg(\forall x \in \mathbb{R}; \ (x^2 > x)) \equiv \exists x \in \mathbb{R}; \ (x^2 \leq x)$

$\neg(\exists x \in \mathbb{R}; \ (x^2 = 2)) \equiv \forall x \in \mathbb{R}; \ (x^2 \neq 2)$

The second original is true — take $x = \sqrt{2}$. So its negation is false. Whether that witness is a fraction is a different question, and it is the last thing we do in Lecture 3.

Example: Negating a Quantified Conditional

Let $P(x)$ and $Q(x)$ be predicates on a domain D. Negate $\forall x \in D;\ P(x) \to Q(x)$.

$$\neg(\forall x \in D;\ P(x) \to Q(x))$$

$$\equiv \exists x \in D;\ \neg(P(x) \to Q(x))$$ FLIP THE QUANTIFIER

$$\equiv \exists x \in D;\ P(x) \land \neg Q(x)$$ NEGATE THE IMPLICATION

In words: to break “every P is a Q,” produce one thing that is a P and is not a Q.

Common Mistakes

01 Negating $\forall x\ P(x)$ as $\forall x\ \neg P(x)$. The quantifier flips: it is $\exists x\ \neg P(x)$.

02 Leaving the domain unstated. $\forall x\ (x^2 \geq x)$ is true over $\mathbb{Z}$ and false over $\mathbb{R}$.

03 Treating $P(x)$ as a proposition. It has no truth value until x is fixed or quantified.

04 Proving $\forall x\ P(x)$ with an example. One example proves $\exists$, and one counterexample refutes $\forall$. Nothing else.

Recap

01 A tautology is always true; $p \equiv q$ means $p \leftrightarrow q$ is one.

02 A chain of named laws beats a table, and keeps working past ten variables.

03 A predicate is a proposition with a hole in it. Fill the hole or quantify it.

04 $\forall$ needs an argument and dies to one counterexample. $\exists$ lives on one witness.

05 Negation flips the quantifier and moves inward: $\neg\forall x\ P(x) \equiv \exists x\ \neg P(x)$.

Next: theorems, lemmas, and four ways to prove one.